

NAME: _____

PERIOD: _____

DATE: _____

What Does This Correlation Support?

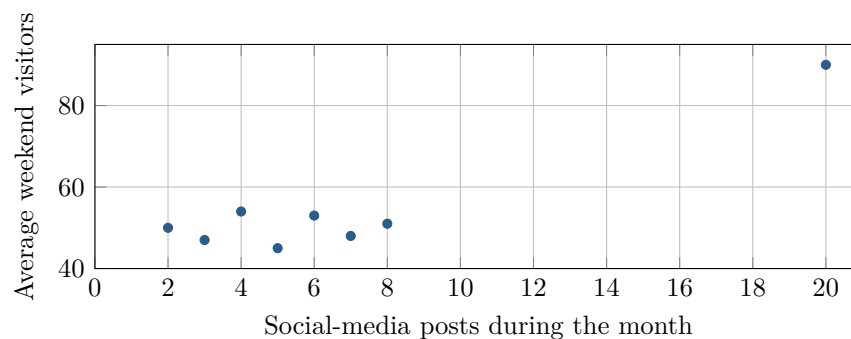
AP Statistics · Unit 5 · Topic 5.2 · B: 2027-02-08 · E: 2027-04-02

[STAR] TODAY'S PROBLEM

Suppose a nonprofit records monthly social-media posts and average weekend visitors at eight community arts centers. The pairs are (2, 50), (3, 47), (4, 54), (5, 45), (6, 53), (7, 48), (8, 51), and (20, 90). For all eight centers, $r = 0.922$.

Omar: Use that strong correlation to forecast the attendance gained from additional posts; every observed center belongs in the analysis.

Tessa: Report the full analysis, but also examine the seven-center cluster and how posting levels were chosen before recommending a campaign.



FIRST TAKE — before the video (5 min)

Which proposal seems more defensible, Omar's or Tessa's? Cite one number, plot feature, or design fact.

[BOOK] READ r WITH THE PLOT AND DESIGN (keep on the page)

Correlation r is unit-free, lies from -1 to 1 , and summarizes direction and strength of linear association. Inspect the scatterplot for form and unusual points. If a separated point changes r greatly, report both the full analysis and a labeled sensitivity analysis. Association in observational data does not establish causation.

Finish after the video:

DOK 1 (a) State the range and units of r , and interpret its sign and magnitude.

DOK 2 (b) Calculate r for all eight centers and for the first seven as a labeled sensitivity analysis. Describe the plot's structure.

DOK 3 (c) Recommend Omar's or Tessa's approach. Reconcile both correlations with the plot, state what the full value supports, use the observational design to assess the forecast, and explain one decision risk from overreaching.

Frame: For all eight centers, $r =$ _____ supports _____, while $r =$ _____ for seven centers shows _____.

Frame: I recommend _____; the design permits _____ but not _____, so _____.

EXIT — turn this in

One thing the video changed about my first take: _____